



E9: 309 Advanced Deep Learning

2-11-2020

Housekeeping

✳ Attendance

✓ We will use the recorded sessions for attendance

★ If you are unable to attend live sessions (due to network or other issues, please indicate by email before or after class to the instructor and copy the FAs).

✳ Mid-term exam

➡ 1st week of Dec. (Modules I and II).



Housekeeping

✳ 1st mini-project

✓ Deadlines

- ★ Abstract submission deadline (Nov 2nd, Monday)
 - ★ Using the google form given in the webpage
- ★ Solo projects or 2-member projects
 - ★ Indicate roles of each member in 2-member project
 - ★ 200 page abstract of the work. If modifications are needed, we will review and let you know in 2-3 days.



Housekeeping

✳ 1st mini-project

✓ Deadlines

- ★ Report and presentation slides (Nov 19, 10 AM).
 - ★ 1-page pdf with second page only for references and tools used (Template will be provided).
- ★ Report - Indicate prior work, technical details and your contribution. Strictly adhere to the guidelines given in the template.
- ★ Slides (max 4 slides) - 4 min presentation for solo project and 6 min. for two member teams. 3 mins for your presentation and 1 min for Q&A.
- ★ Two slots are available on 2 days (pick the suitable based on your other class schedules).



Recap of previous class



State of affairs

- ✳ Encoder-decoder models with attention.

 - ➡ self attention and multi-head attention

- ✳ Transformer models - Introduction

Transformers

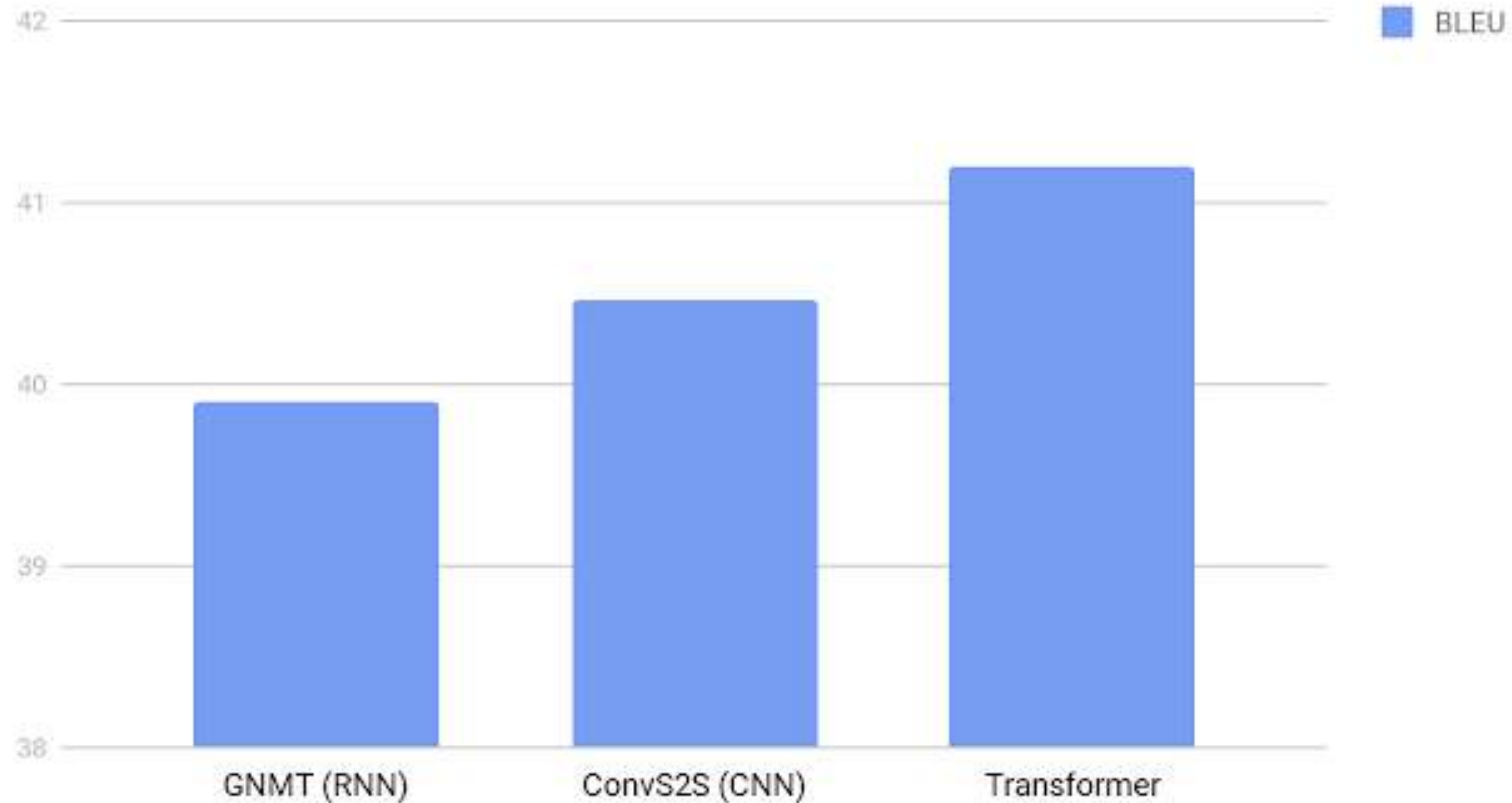
- * Encoder Decoder architecture based models.
- * Uses only feed forward architectures with self-attention.
- ➔ Multi-head self attention.
- * All the encoder layers and the decoder layers have the same set of operations.

Reading Assignment - “Attention is All You Need”
<https://arxiv.org/pdf/1706.03762.pdf>



Transformers - the state of art in NMT

English French Translation Quality



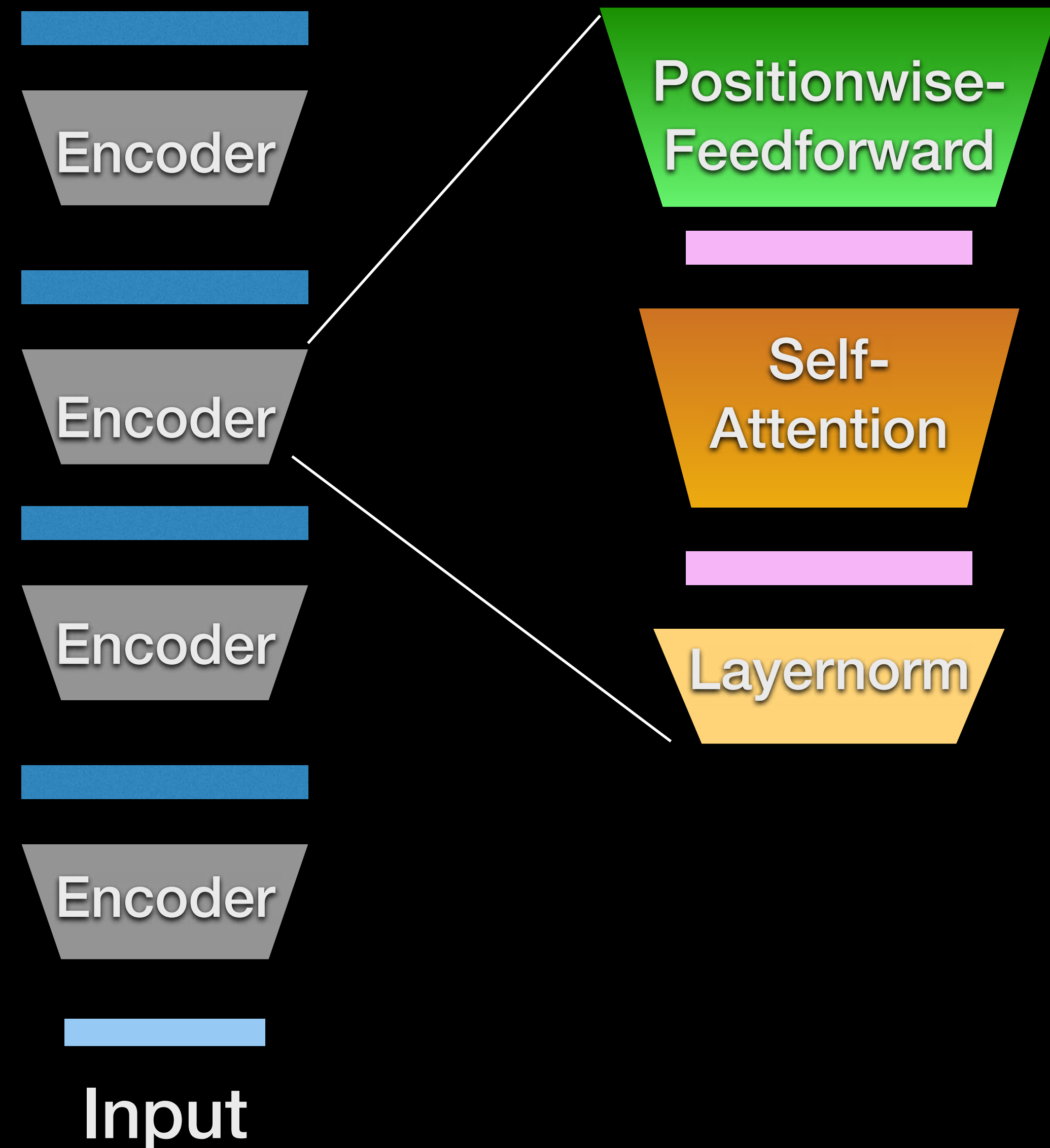
Transformers - the state of art in NMT



Transformers

✳ Encoder layers

- ➔ Consist of layer norm
- ➔ Self attention (multi-head)
- ➔ Positionwise feedforward
- ✓ May also consist of skip connections.



Transformers - encoder

✳ Let $\mathbf{x}(1) \dots \mathbf{x}(T)$ denote the input and let $\mathbf{e}^l(1) \dots \mathbf{e}^l(T)$ denote encoder outputs at layer l .

$$\overline{\mathbf{E}}^{l-1} = \text{Layernorm}([\mathbf{e}^{l-1}(1) \dots \mathbf{e}^{l-1}(T)]^T) \in \mathcal{R}^{T \times D}$$

✳ Definition of layer norm

$$\text{Layernorm}(\mathbf{e}^l(t)) = \frac{\alpha^l}{\sigma_{\mathbf{e}^l(t)}} \odot (\mathbf{e}^l(t) - \mu_{\mathbf{e}^l(t)}) + \beta^l$$



Transformers - encoder

Query, Key and Value

encoder layer

$T \times H$

$$\vec{Q}_h^l = \vec{E}^{l-1} \mathbf{W}_h^{l,Q} + \mathbf{1}(\mathbf{b}_h^{l,Q})^T \in \mathcal{R}^{T \times d}$$

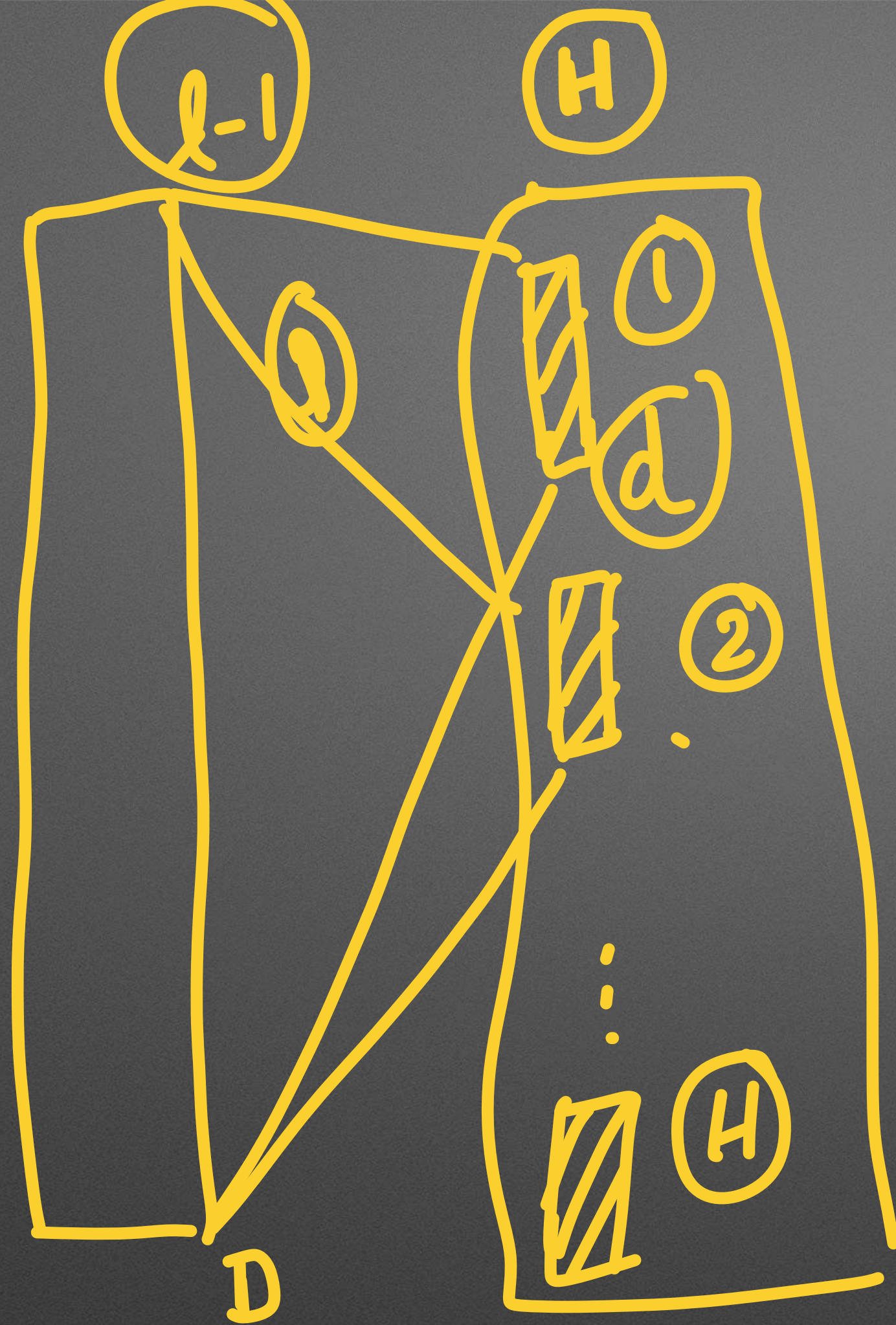
$$\vec{K}_h^l = \vec{E}^{l-1} \mathbf{W}_h^{l,K} + \mathbf{1}(\mathbf{b}_h^{l,K})^T \in \mathcal{R}^{T \times d}$$

$$\vec{V}_h^l = \vec{E}^{l-1} \mathbf{W}_h^{l,V} + \mathbf{1}(\mathbf{b}_h^{l,V})^T \in \mathcal{R}^{T \times d}$$

$$\mathbf{W}_h^{l,Q}, \mathbf{W}_h^{l,K}, \mathbf{W}_h^{l,V} \in \mathcal{R}^{D \times d} \quad \mathbf{b}_h^{l,Q}, \mathbf{b}_h^{l,K}, \mathbf{b}_h^{l,V} \in \mathcal{R}^{d \times 1}$$

$$h = \{1..H\} \text{ heads} \quad d = \frac{D}{H} \quad \mathbf{1} \in \mathcal{R}^{T \times 1} \text{ all ones}$$





$$l \rightarrow \underline{\underline{D = d \times H}}$$

Transformers - encoder

$T = 100$

* Multi-head attention

$$\hat{\mathbf{A}}_h^l = \underbrace{\mathbf{Q}_h^l}_{1 \times d} (\underbrace{\mathbf{K}_h^l}_{1 \times d})^T \in \mathcal{R}^{T \times T}$$

$$\tilde{\mathbf{A}}_h^l = \text{softmax}\left(\frac{\hat{\mathbf{A}}_h^l}{\sqrt{d}}\right)$$

$\mathcal{R}^{T \times T}$

$$\left(\mathbf{A}_h^l \perp \mathbf{A}_{h+1}^l \right) \\ \mathbf{A}_h^l \mathbf{A}_{h+1}^{lT} \approx \mathbf{I}$$

$$\mathbf{C}_h^l = \mathbf{A}_h^l \mathbf{V}_h^l \in \mathcal{R}^{T \times d}$$

* Context vector from self-attention

$$\mathbf{C}^l = [\mathbf{C}_1^l \dots \mathbf{C}_H^l] \in \mathcal{R}^{T \times D}$$



Transformer - encoder

✳ Position wise feedforward layer



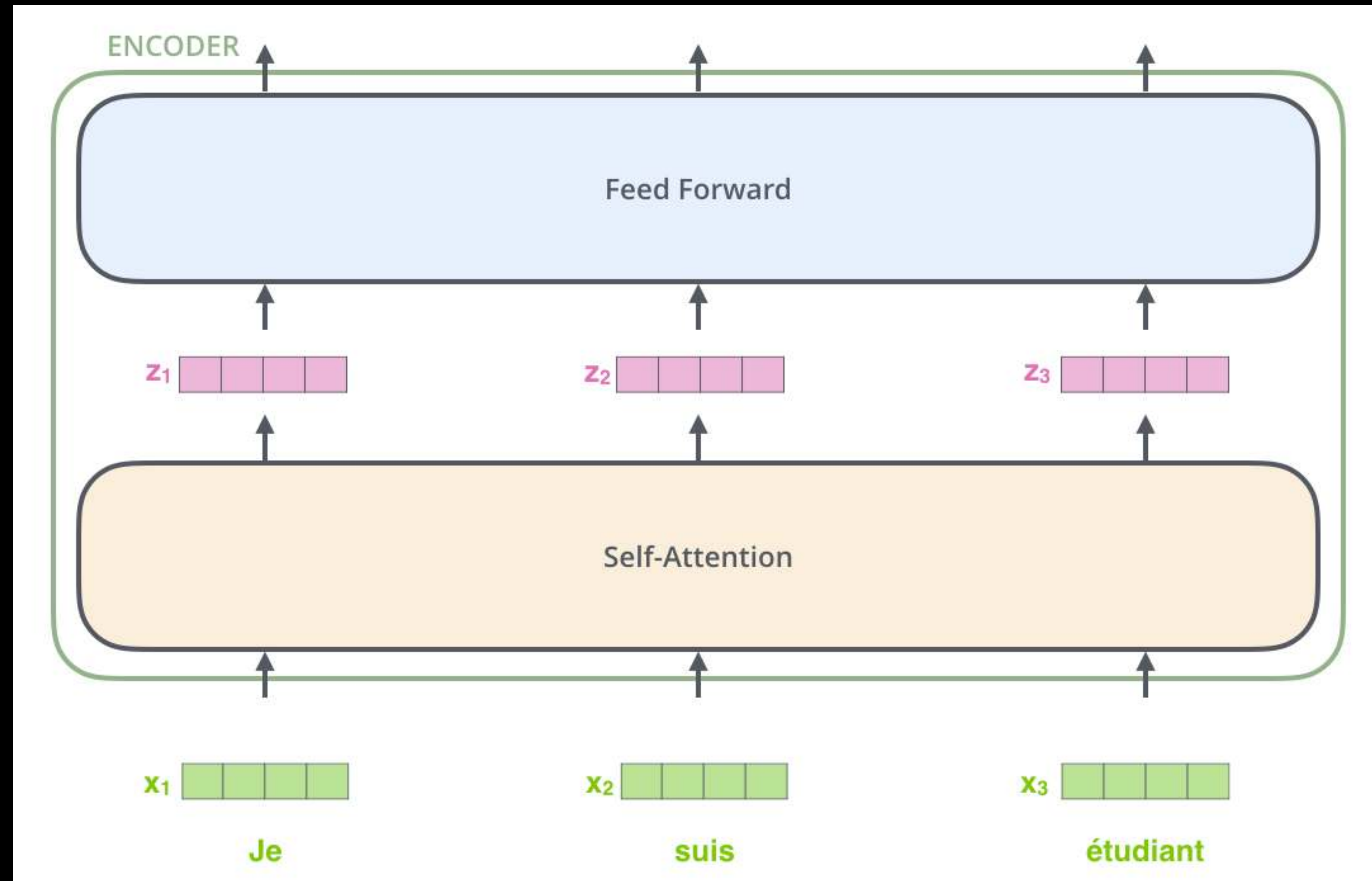
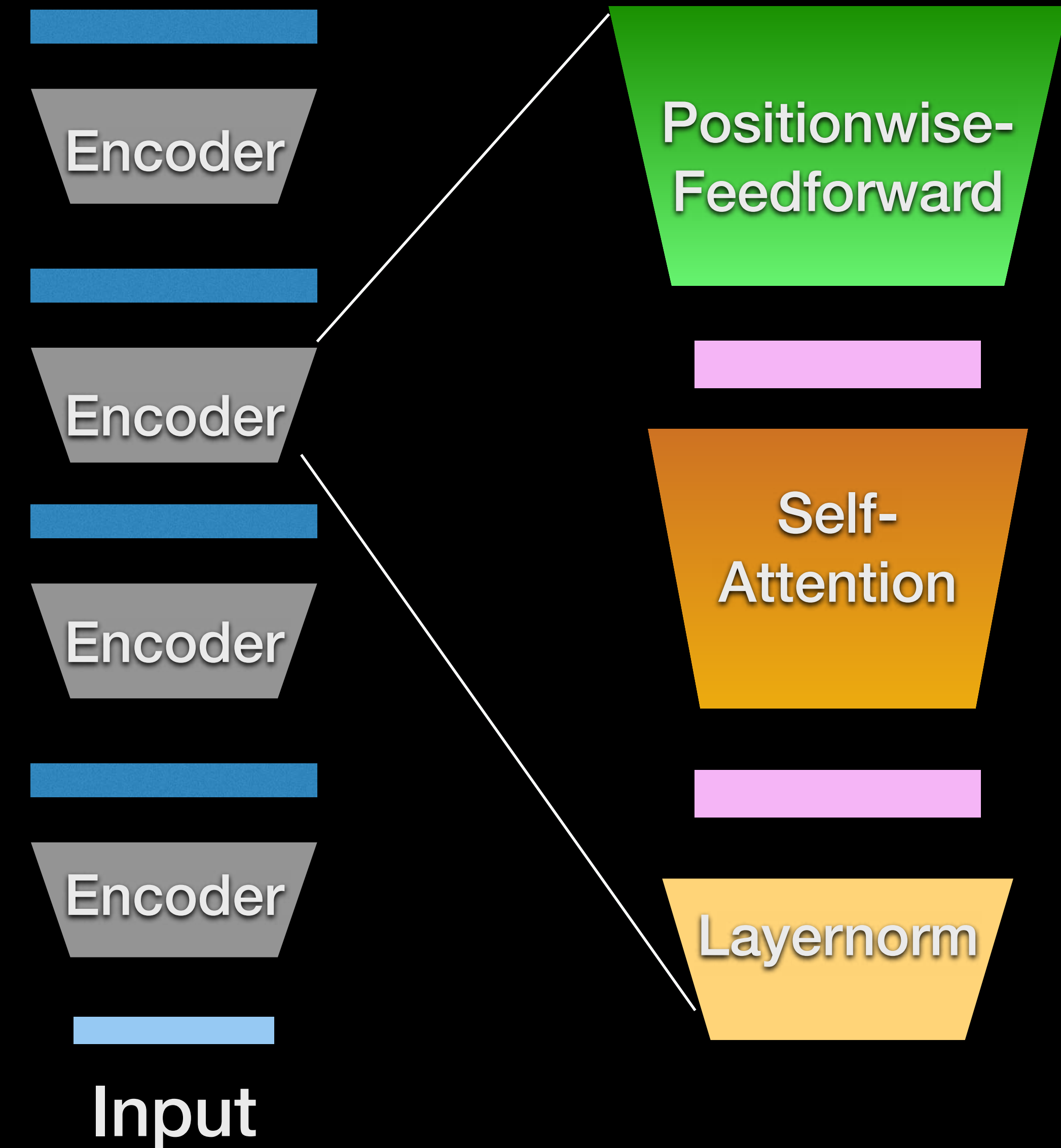
$$\mathbf{E}^l_{ff} = \text{ReLU}(\mathbf{C}^l \mathbf{W}^l_{ff} + \mathbf{1} \mathbf{b}^T_{ff}) \in \mathcal{R}^{T \times d_{ff}}$$

✳ Encoder layer output

$$[\mathbf{e}^l(1) \dots \mathbf{e}^l(T)] = \mathbf{E}^l_{ff} \mathbf{W}^l_{of} + \mathbf{1} (\mathbf{b}^l_{of})^T \in \mathcal{R}^{T \times D}$$

↑ ↑

Transformers - encoder



Self Attention - recap

10.1 10.0 9.9
1/3 1/3 1/3 → 2/3 (1-2/3)
1.01 1.0 0.99

* Illustrative example - The quick brown fox (English) —> Der shnelle brane fuchs (German)

1 2 3 4
The quick brown fox

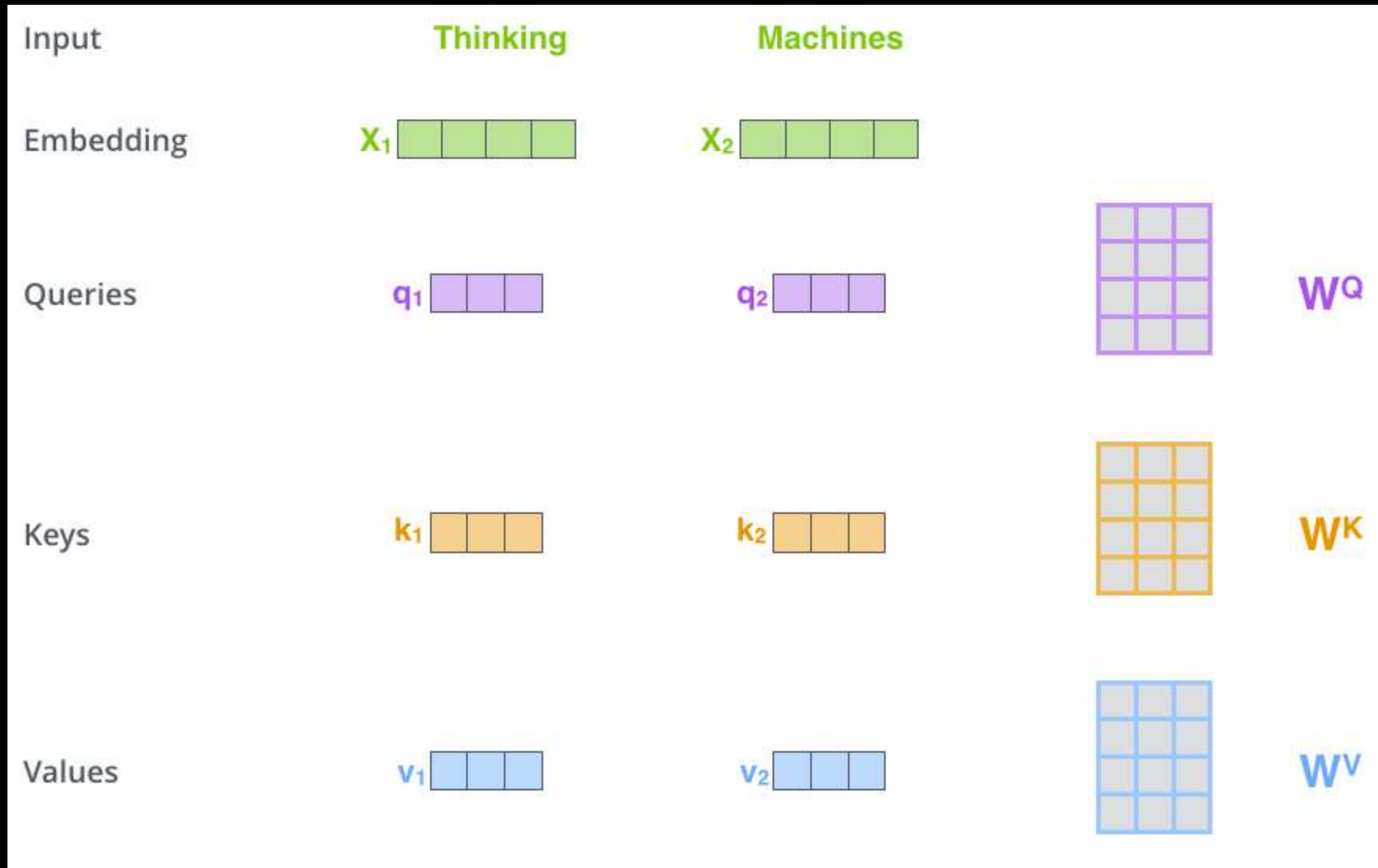
	THE	QUICK	BROWN	FOX
THE	0.9	0.1	0	0
QUICK	0.1	0.75	0	0.15
BROWN	0	0	0.7	0.3
FOX	0	0.1	0.35	0.55

T = 4
 $A^l_h \in \mathbb{R}^{4 \times 4}$
↑ softmax
softmax($\frac{QK^T}{\sqrt{d}}$)

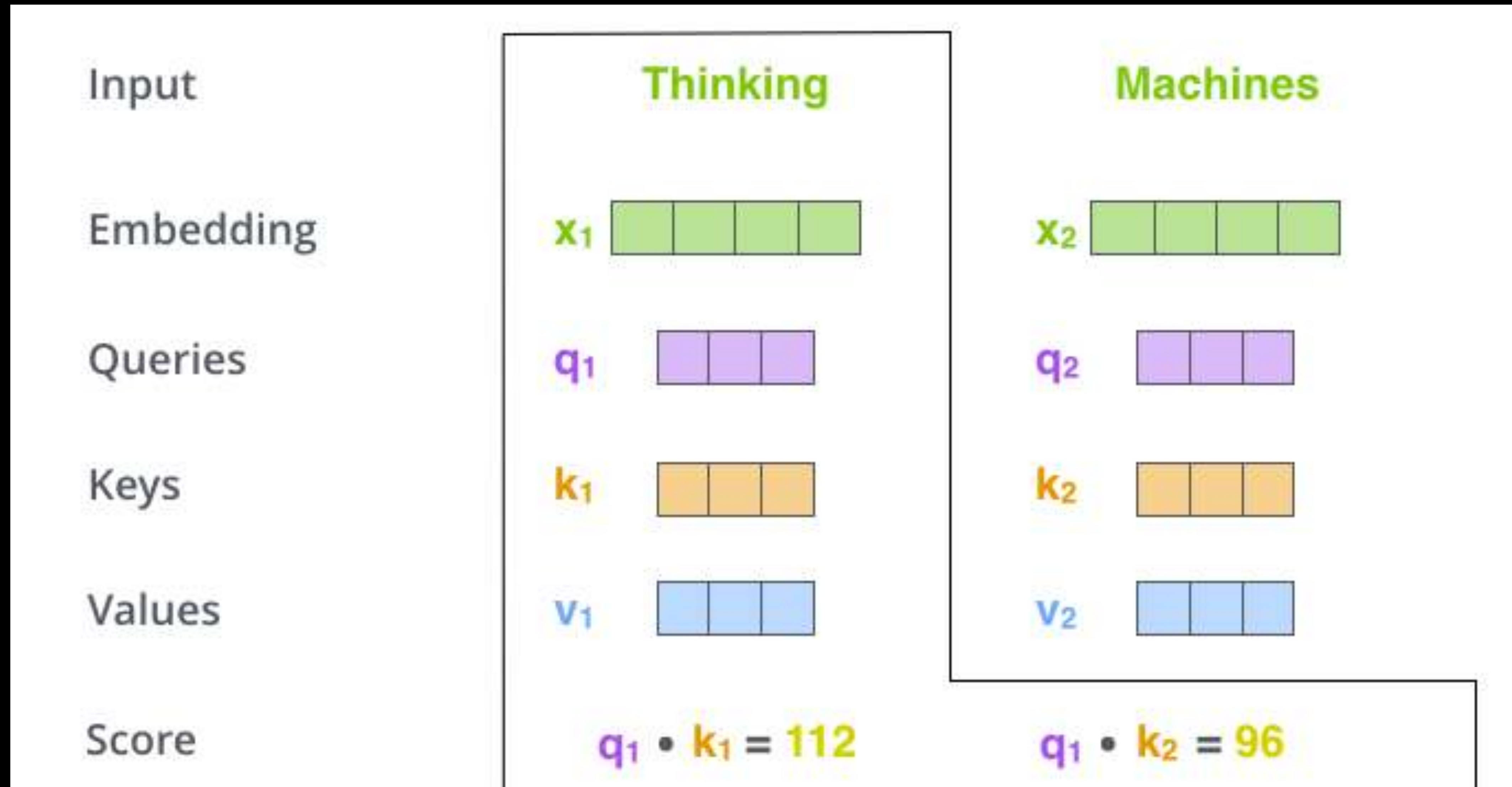
→
→
→
→



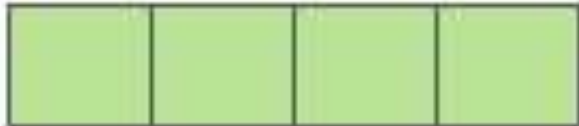
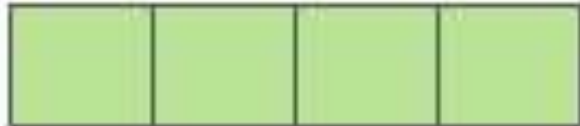
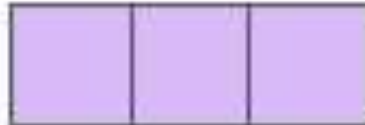
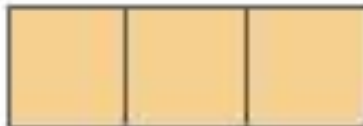
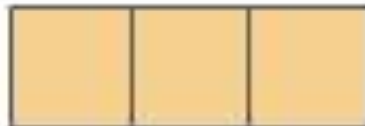
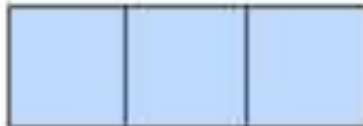
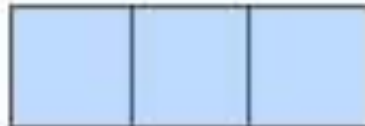
Self-attention revisited



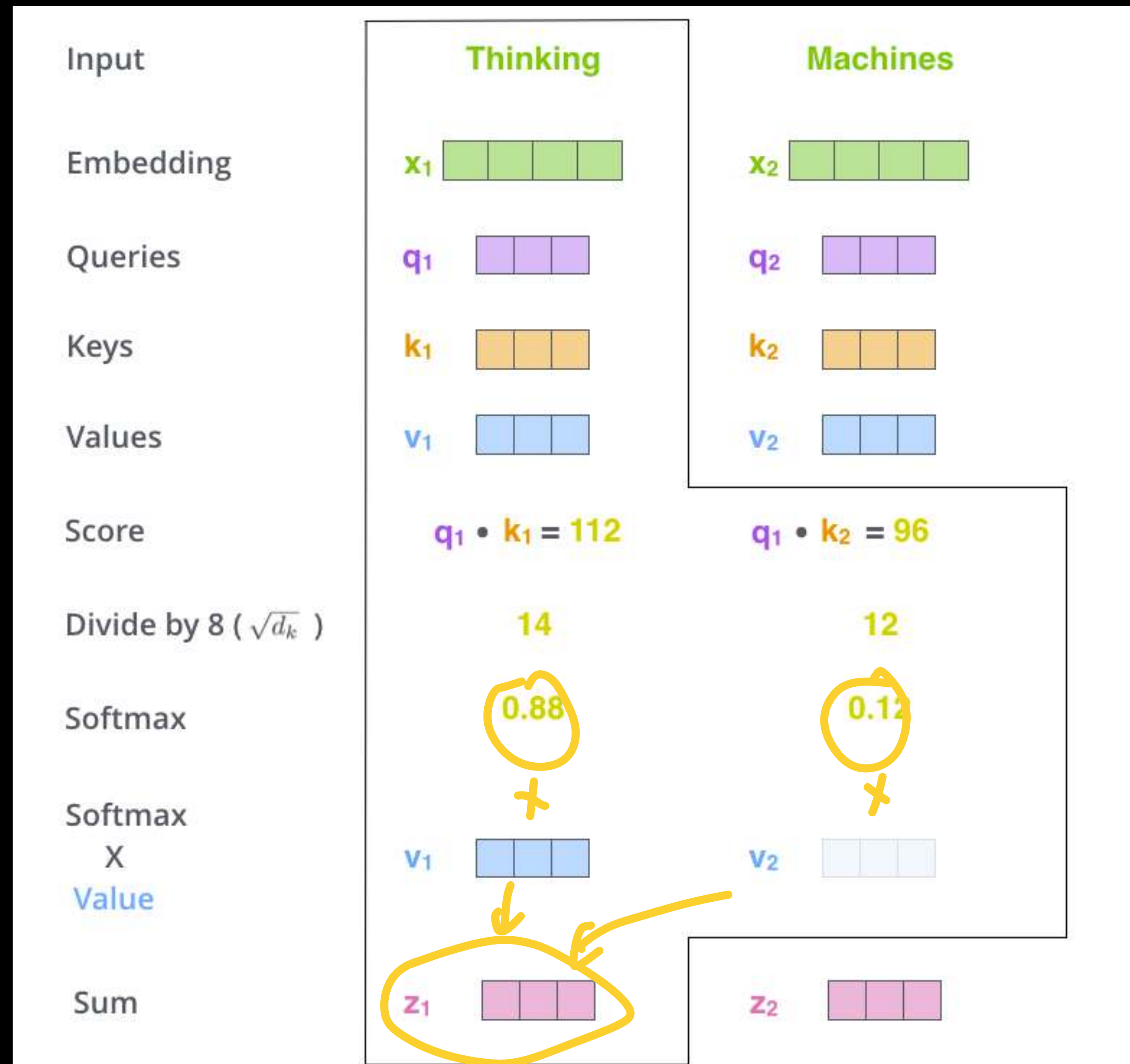
Self-attention revisited



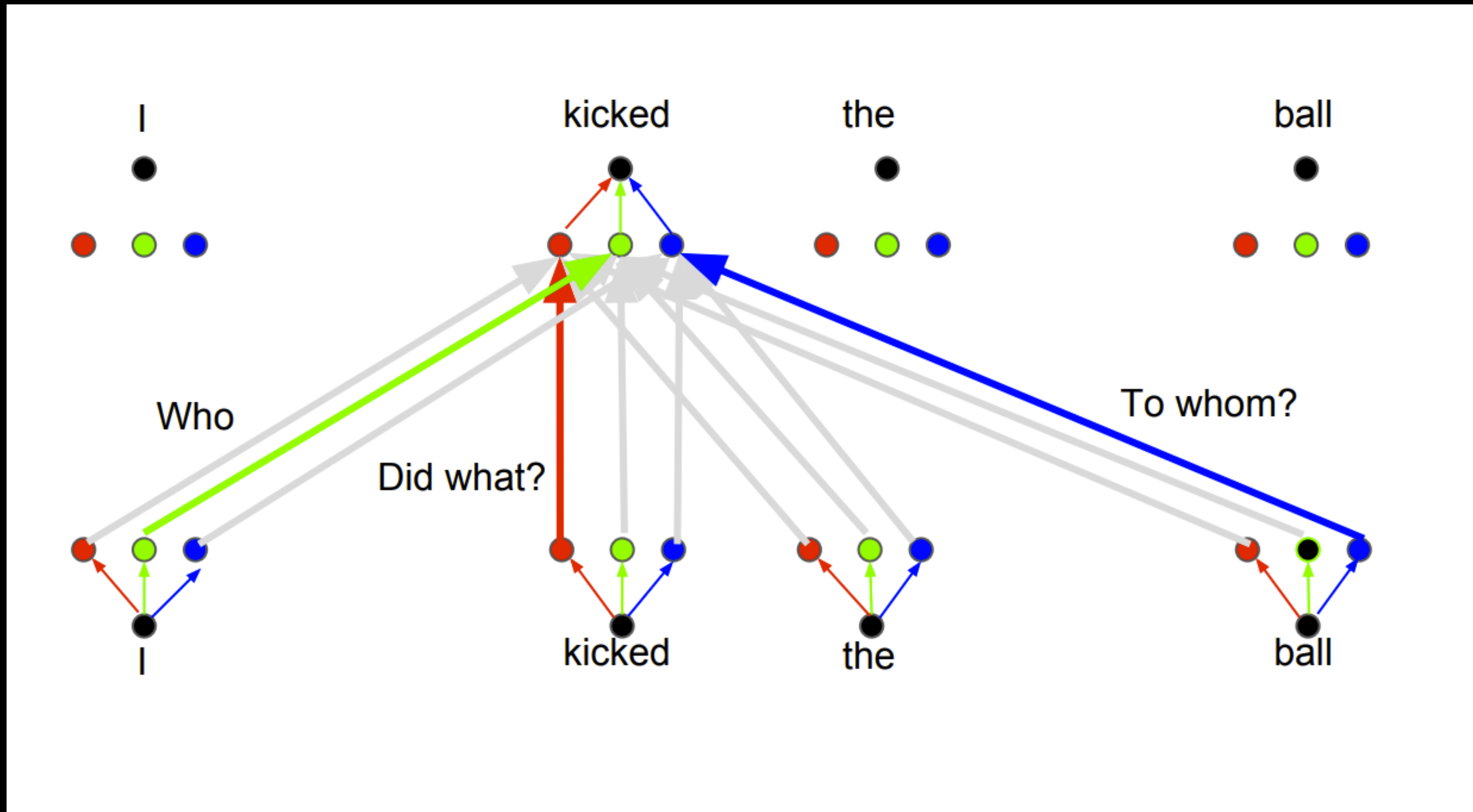
Self-attention revisited

Input	Thinking	Machines
Embedding	x_1 	x_2 
Queries	q_1 	q_2 
Keys	k_1 	k_2 
Values	v_1 	v_2 
Score	$q_1 \cdot k_1 = 112$	$q_1 \cdot k_2 = 96$
Divide by 8 ($\sqrt{d_k}$)	14	12
Softmax	0.88	0.12

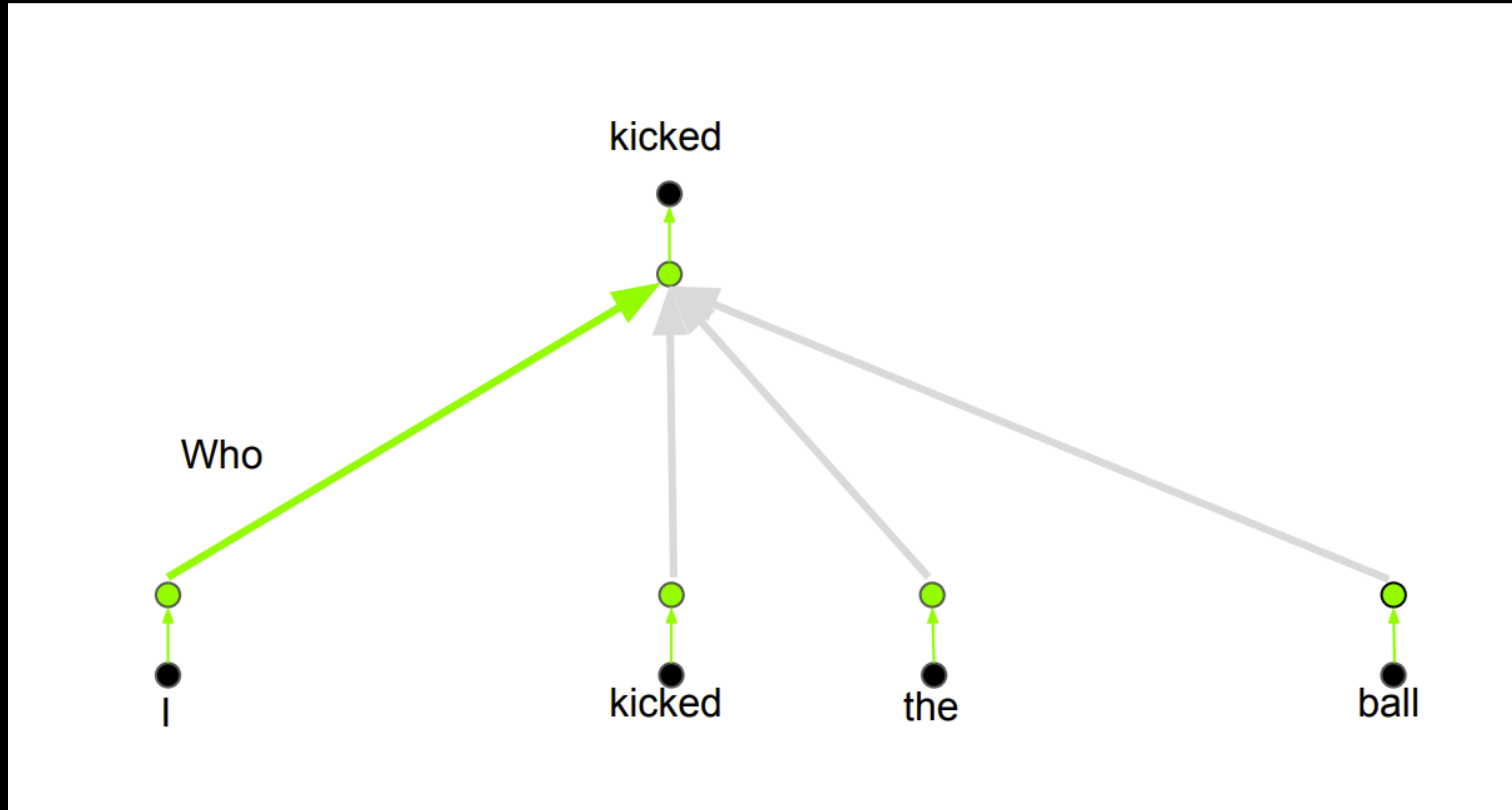
Self-attention revisited



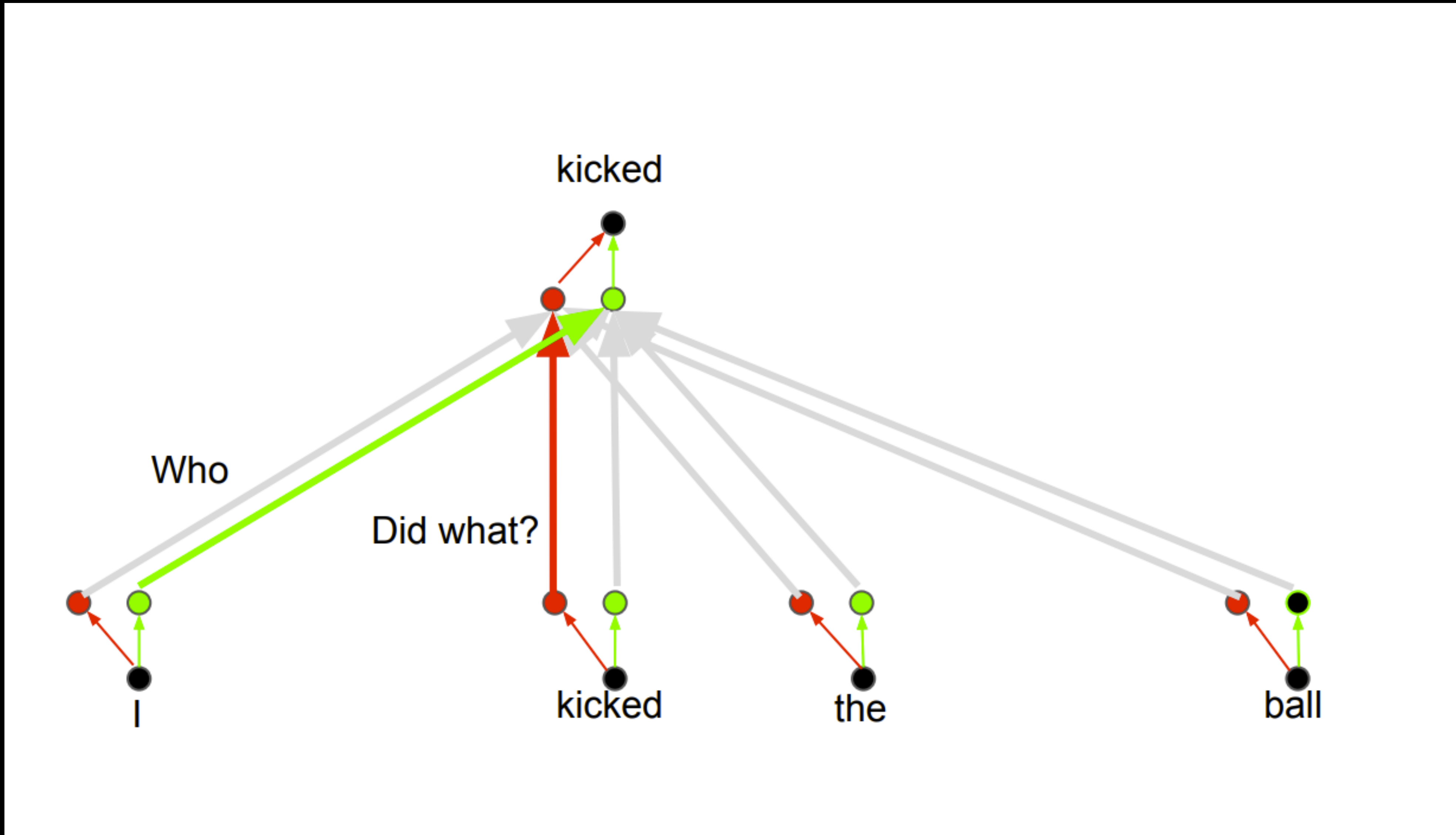
Self-attention multi-head



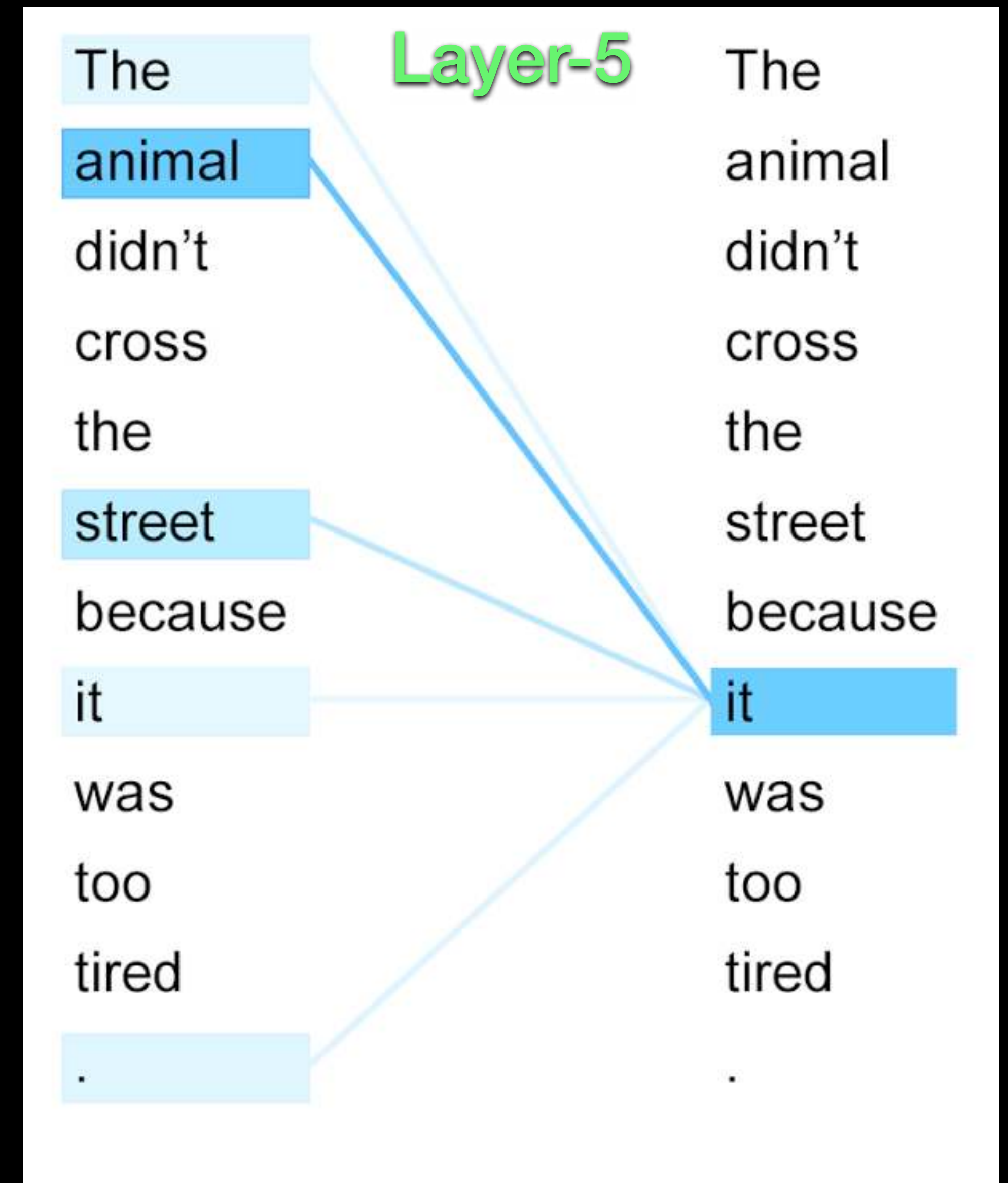
Self-attention multi-head - role of attention heads



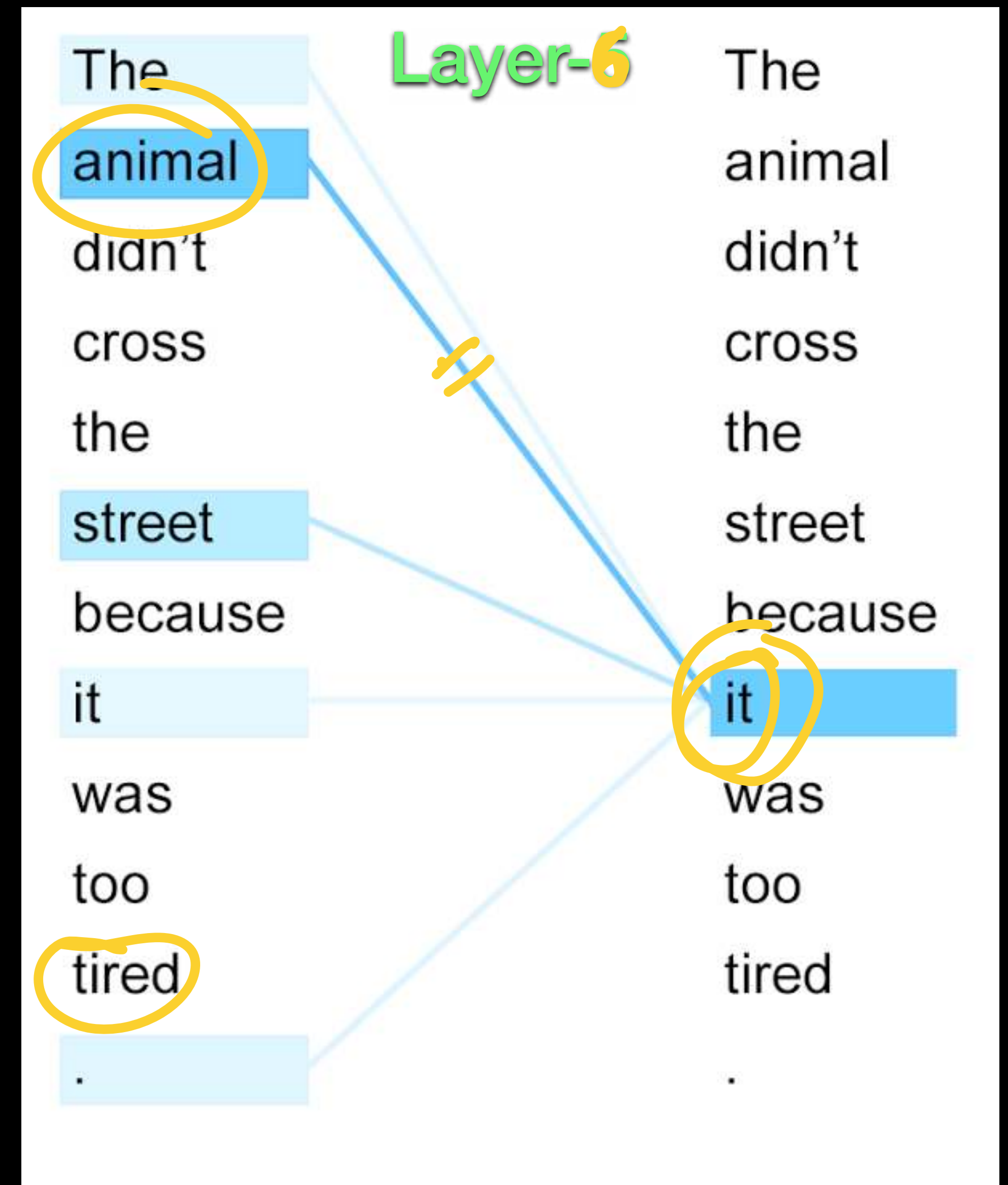
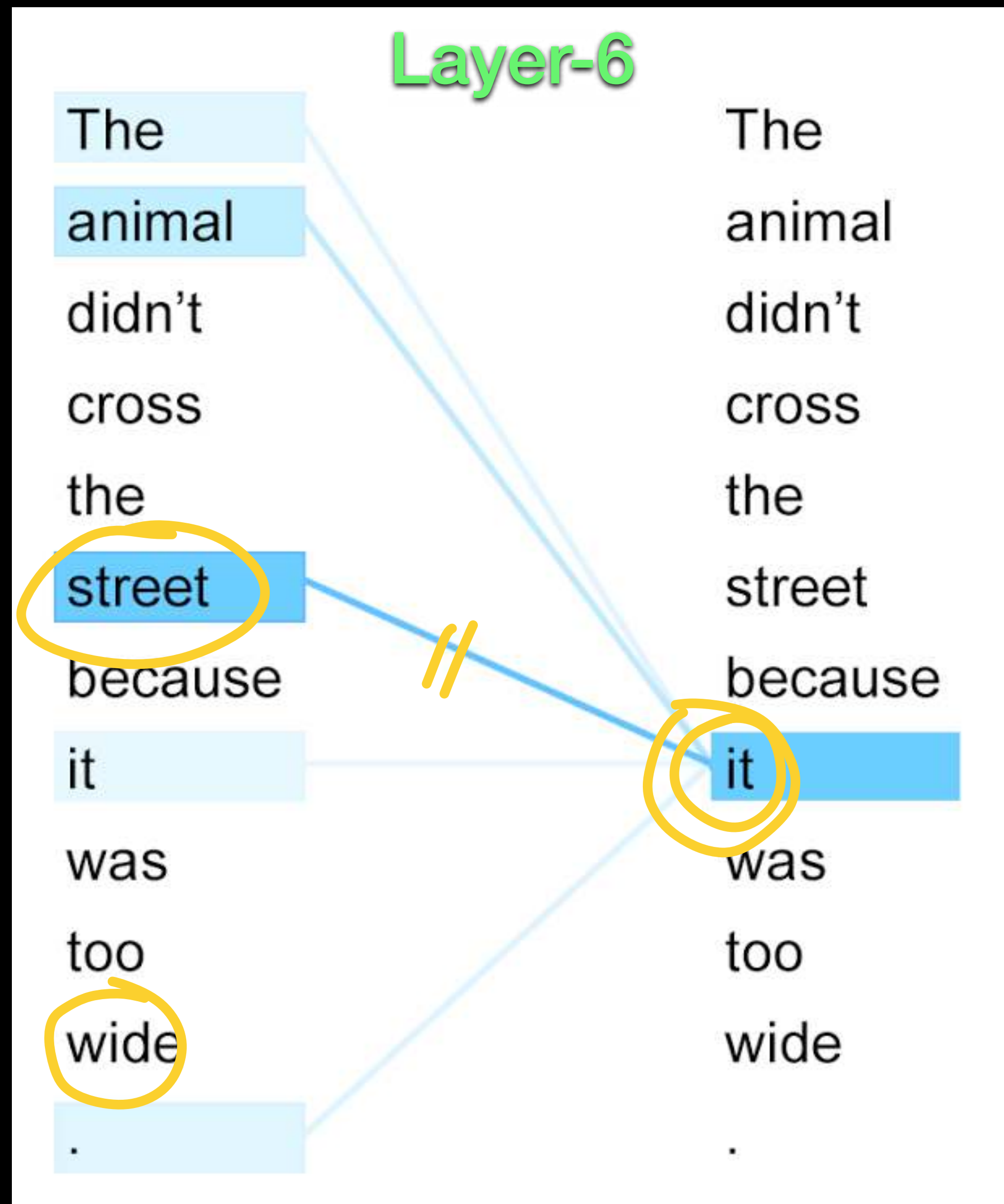
Self-attention multi-head - role of attention heads



Self-attention - need for depth

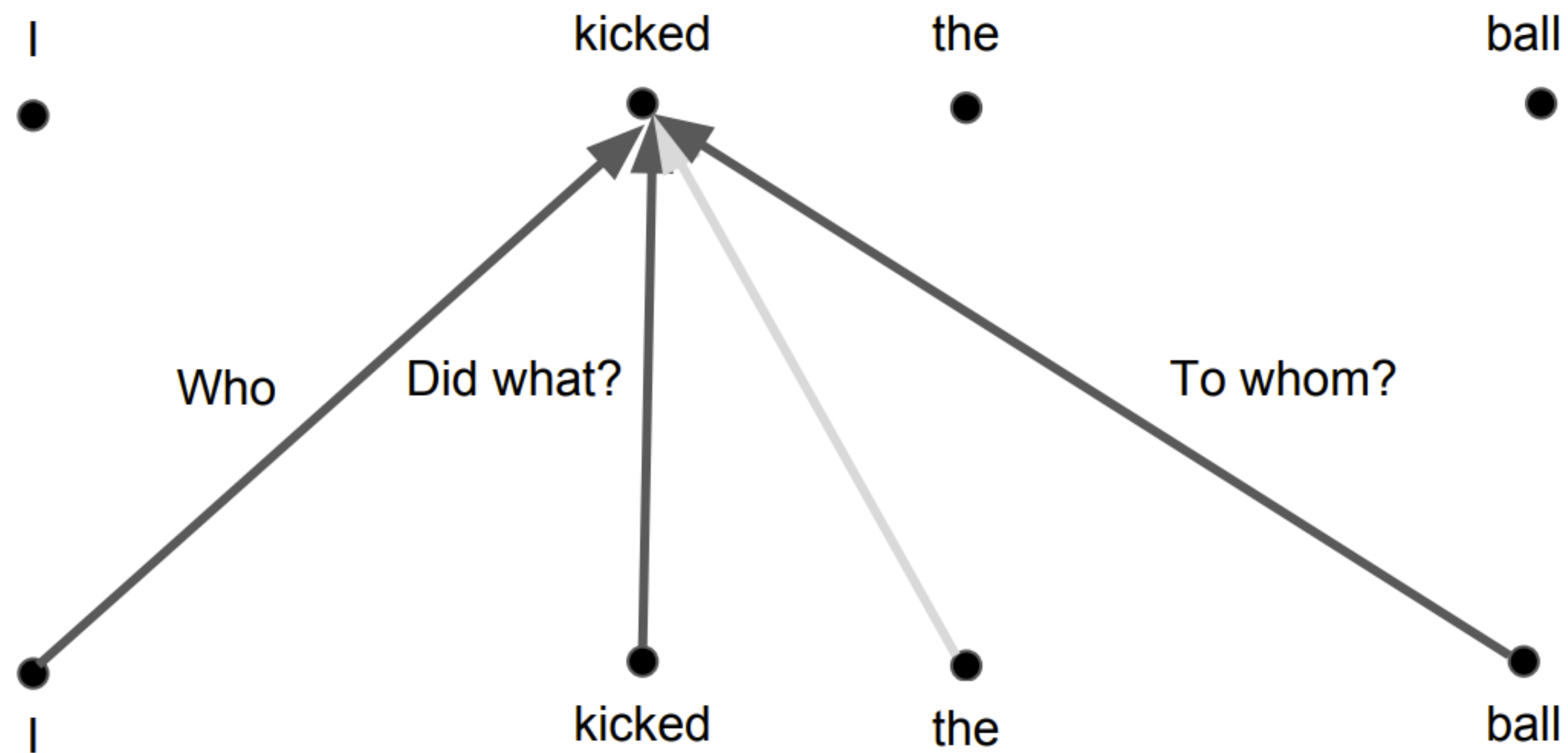


Self-attention - need for depth



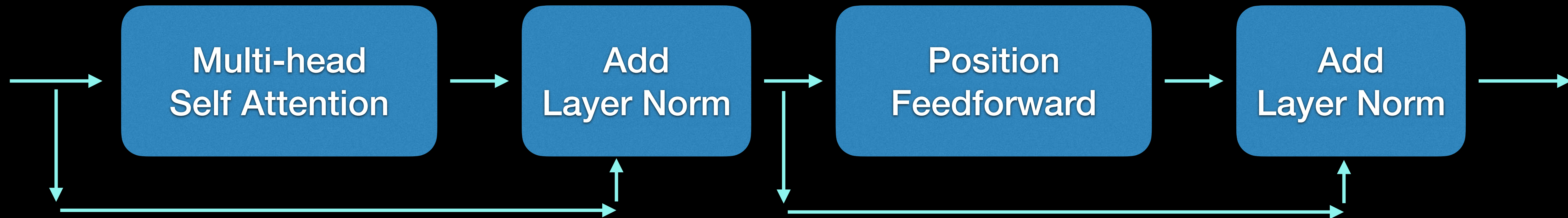
Need for multi-head attention

Self-Attention



Single layer of encoder (typical implementation)

- * Single encoder layer has typically self-attention skip connection, layer norm and feedforward layer



Positional encoding

✳ No recurrence or position awareness yet in the model

Binary format -

position can encode the rate of
change of bits across time

In floating format - one can use
sines and cosines

0 :	0	0	0	0	8 :	1	0	0	0
1 :	0	0	0	1	9 :	1	0	0	1
2 :	0	0	1	0	10 :	1	0	1	0
3 :	0	0	1	1	11 :	1	0	1	1
4 :	0	1	0	0	12 :	1	1	0	0
5 :	0	1	0	1	13 :	1	1	0	1
6 :	0	1	1	0	14 :	1	1	1	0
7 :	0	1	1	1	15 :	1	1	1	1



Positional encoding

✳ An example used in the first paper

$$\mathbf{p}(t) \in \mathcal{R}^D$$

$$p_i(t) = \begin{cases} \sin(\omega_k t), & \text{if } i = 2k \\ \cos(\omega_k t), & \text{if } i = 2k + 1 \end{cases} \quad k \in \{1 \dots \frac{D}{2}\}$$

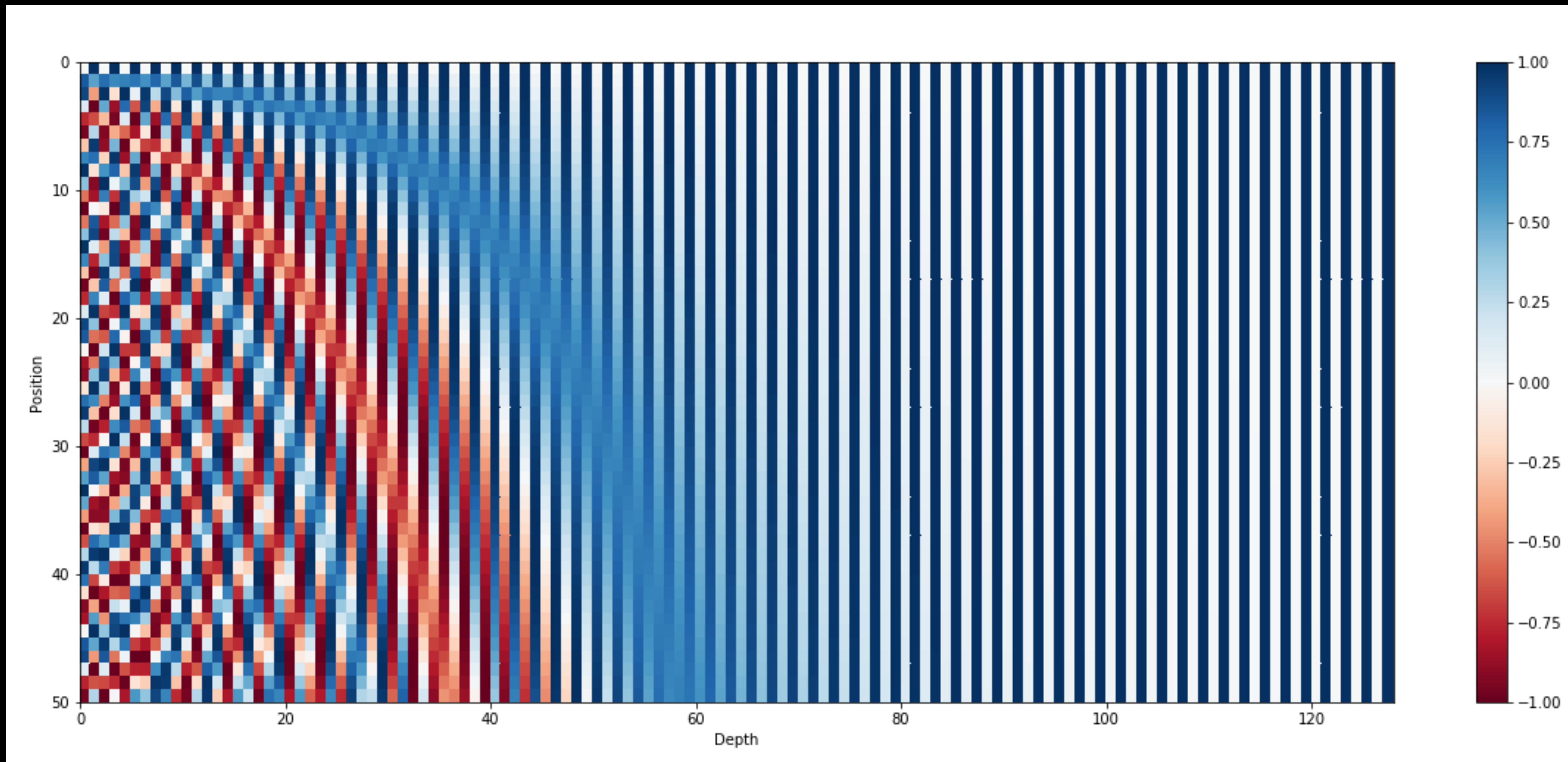
$$\omega_k = \frac{1}{10000^{\frac{2k}{D}}}$$

$$\mathbf{x}(t) = \mathbf{x}(t) + \mathbf{p}(t)$$



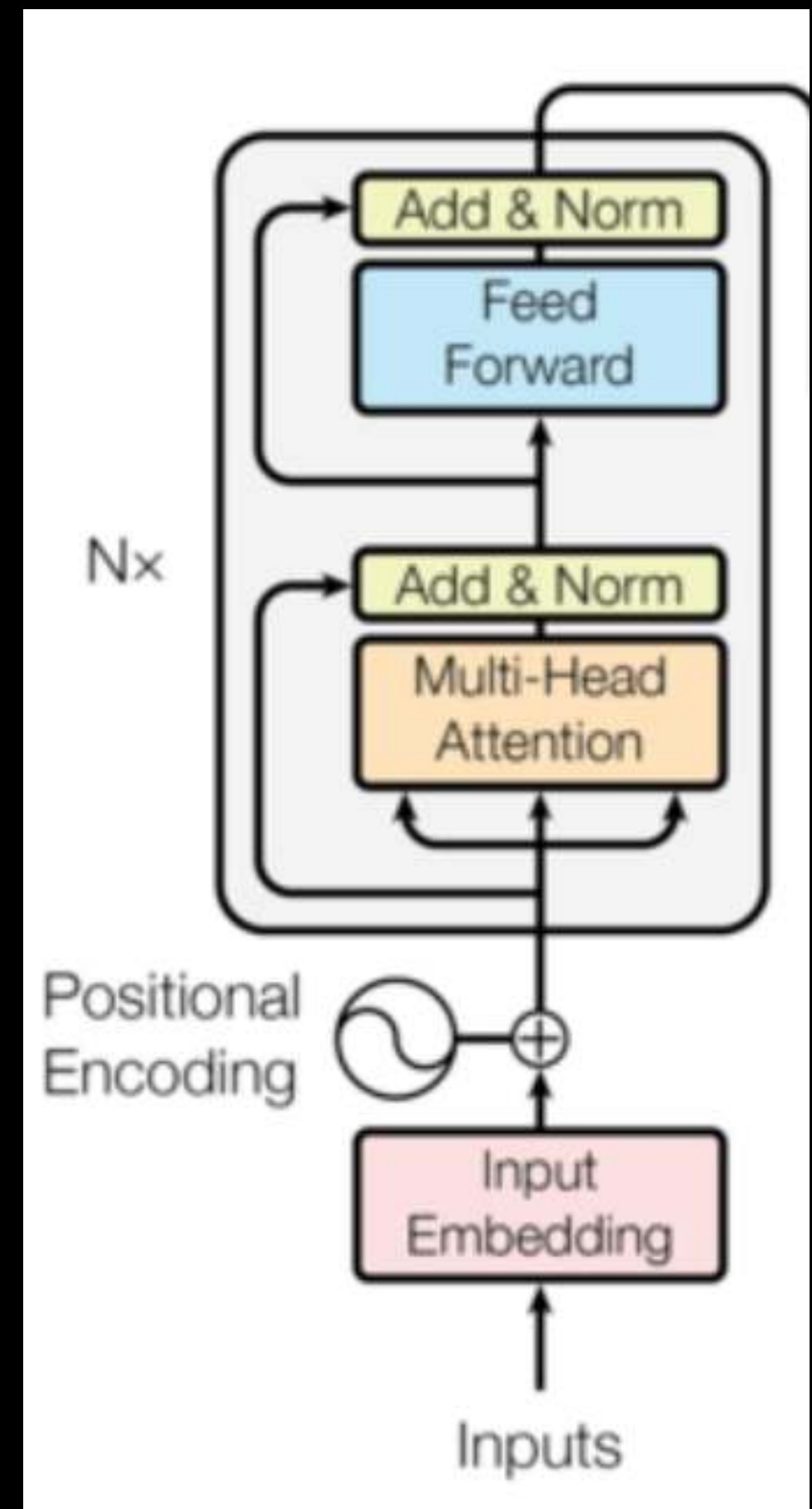
Positional encoding

✳ An example used in the first paper $\mathbf{p}(t) \in \mathcal{R}^D$ [T=50, D=128]



Transformer encoder - overview

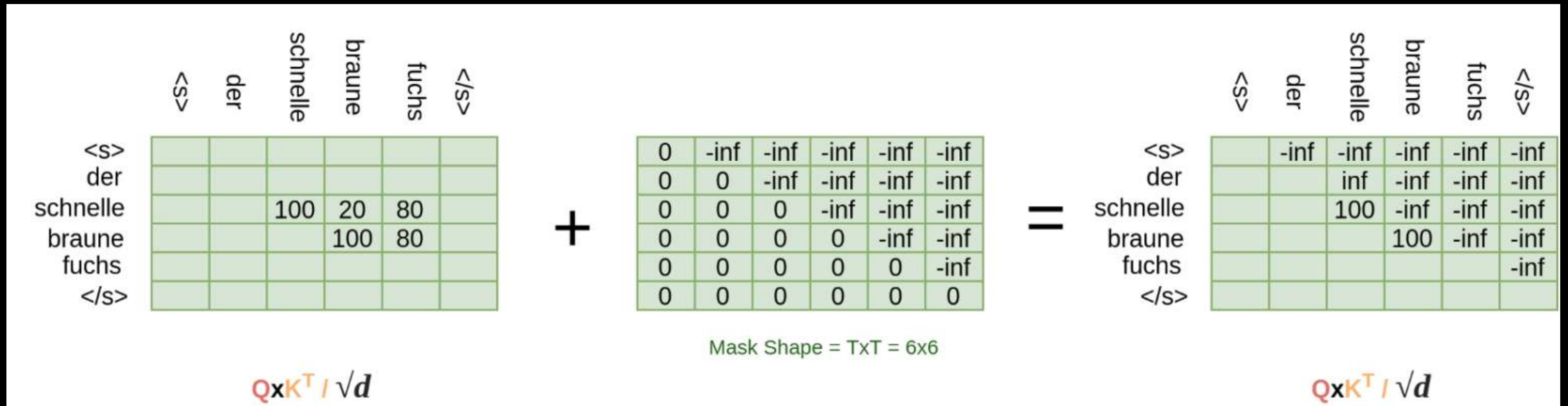
Reading Assignment - “Attention is All You Need”
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Transformer decoder

✳ Masked self-attention layer -

- ✓ Mask makes the output dependencies causal
- ★ Only the past is used to encode the attention.



Transformer decoder

✳ Masked self-attention layer -

- ✓ Mask makes the output dependencies causal
- ★ Only the past is used to encode the attention.

$$\text{Softmax}\left\{\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right\}\mathbf{V} \longrightarrow \text{Softmax}\left\{\text{Mask} + \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right\}\mathbf{V}$$

- ★ Make the attention matrix to be lower triangular



Transformer decoder

✳ Masked self-attention layer -

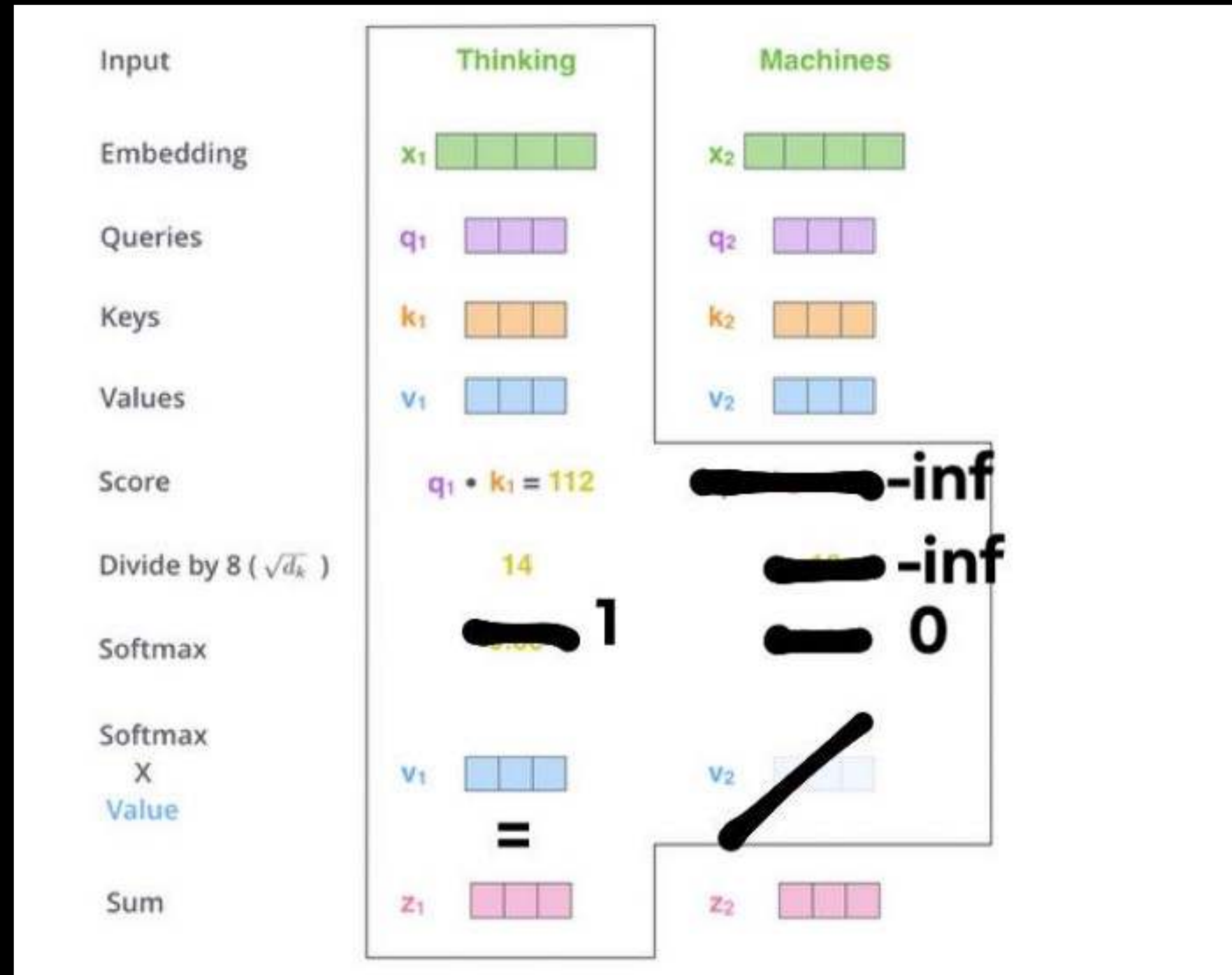
- ✓ Mask makes the output dependencies causal
- ★ Only the past is used to encode the attention.

	<s>	der	schnelle	braune	fuchs	</s>
<s>		0	0	0	0	0
der			0	0	0	0
schnelle			.75	0	0	0
braune				.85	0	0
fuchs						0
</s>						

$$\text{softmax}(QxK^T / \sqrt{d})$$



Transformer decoder



Encoder-decoder attention

✳ Use the Key and Value matrices from the last layer of the encoder

$$\mathbf{Q}_h^p = \overline{\mathbf{D}}^{p-1} \mathbf{W}_h^{p,Q} + \mathbf{1}(\mathbf{b}_h^{p,Q})^T \in \mathcal{R}^{S \times d}$$

$$\mathbf{K}_h^p = \mathbf{E}^L \mathbf{W}_h^{p,K} + \mathbf{1}(\mathbf{b}_h^{p,K})^T \in \mathcal{R}^{T \times d}$$

$$\mathbf{V}_h^p = \mathbf{E}^L \mathbf{W}_h^{p,V} + \mathbf{1}(\mathbf{b}_h^{p,V})^T \in \mathcal{R}^{T \times d}$$

$$\mathbf{D}_h^p = \text{softmax}\left(\frac{\mathbf{Q}_h^p (\mathbf{K}_h^p)^T}{\sqrt{d}}\right) \mathbf{V}_h^p \in \mathcal{R}^{S \times d}$$

$$h = \{1..H\} \quad \text{heads} \quad d = \frac{D}{H}$$



Transformer - decoder

✳ Decoder Layer Output

✳
$$[\mathbf{d}^p(1) \dots \mathbf{d}^p(S)] = \text{ReLU} \left(\mathbf{D}_{ff}^p \mathbf{W}_{of}^p + \mathbf{1} (\mathbf{b}_{of}^p)^T \right) \in \mathcal{R}^{S \times D}$$

Transformer - full pipeline

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